# PHILIP UMEADI

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## EDUCATION

**Syracuse University** Syracuse, NY
College of Arts & Sciences August 2023 – May 2026
Bachelor of Science in Applied Mathematics, Minor in Economics Dean's List (Fall 2025)

## RELEVANT WORK EXPERIENCE

**Verisk** | Analyst Intern | Jersey City, NJ June 2025 – August 2025

- Collaborated with senior developers to engineer a scalable Python framework to complete a vendor record migration of 27,000+ legacy documents to a new database ahead of annex season, when contracts are reviewed for renegotiation, leveraging pandas vectorization, regex, and RapidFuzz
- Engineered a programmatic automation suite to replace manual mail-merge workflows.

**Extraordinary Re** | Research Intern | New York, NY May 2023 – August 2023

- Researched how a consortium model could attract institutional participants and provide liquidity for liquid insurance contract markets and recruited a former GE Private Equity VP

## RESEARCH & PROJECTS

**Mathematical Foundations of Machine Learning & NLP**, **supervised by Justin Ko, PhD | Syracuse University**

- Conducting a structured review of MML (Deisenroth et al.), ISL (James et al.), and SLP (Jurafsky & Martin), progressing from linear regression, PCA, and kernel-based SVMs to deep learning. This includes studying the mathematics of neural network training with gradient descent, network architectures, and regularization, as well as implementing selected topics in Python.
- Developed a nonlinear regression model in Python to forecast natural gas prices, using polynomial-sinusoidal basis functions and Maximum Likelihood Estimation. The model captured seasonal trends from 48 months of historical data to project prices one year ahead. Optimized model complexity through RMSE-based selection to prevent overfitting and ensure robustness.
- **Character-Level N-gram Language Models for Hangman:** Built a character-level n-gram language model in Python to play Hangman on a 370K-word corpus; combined unigram through trigram models via linear interpolation with perplexity-optimized mixture weights, achieving 8.2 average incorrect guesses per word. Showed that the interpolation framework self-regulates model complexity based on data availability. Paper and code available here: https://github.com/mindphil/mml/tree/main/hangman

**Binary Fission as a Homogeneous Branching Process, MAT 526 Independent Project**

- Modeled bacterial colony extinction under antibiotic treatment as a homogeneous branching process. Derived a closed-form threshold for the minimum drug concentration guaranteeing eradication, then compared the theoretical efficacy of two real antibiotics using pharmacodynamic parameters from the literature (Bogdanov et al.). Built a Python/Manim simulation that animates the stochastic process. Paper and code available here: https://github.com/mindphil/branching-process-fission

**Combinatorial Game Theory & C++ Systems Programming**

- Investigated Kimberly Wood's n-cop generalization of the Shannon Switching Game on complete graphs and conjectured that the threshold grows linearly rather than the quadratic bound previously established; presented at a seminar (Jan. 2026).
- Currently building low-level C++ proficiency through structured challenges in bit manipulation and bitboard representation, with the goal of interpreting and contributing to a collaborator's solver

**Linear Algebra and Category Theory, supervised by Ben Kaufman | Syracuse University**

- Applied Category Theory to matrix classification, studying the Kronecker problem and the decomposition of complex linear systems into indecomposable structures

## TECHNICAL SKILLS

**Programming:** Python (pandas, NumPy), C++ (basic), MATLAB (Numerical Methods)
**Tools:** Git, Bloomberg Terminal, Excel/VBA
**Mathematics:** Probability theory, stochastic processes, numerical methods, (applied) linear algebra, ODEs & PDEs

## LEADERSHIP & ACTIVITIES

**Syracuse University Division 1 Rowing** August 2023 – Present

- 20+ hours of weekly training

**Manhattan College FED Challenge** September 2022 – May 2023

- Built econometric models forecasting CPI, unemployment, and GDP

**Manhattan College Investment Club** September 2022 – May 2023

- Conducted DCF and comparable company analysis on equity positions for student-managed portfolio